

Operators, Quantifiers, and NP-Internal Structure: An Expanded Cognitive and Cross-Linguistic Account of Nominal Architecture

By Maks Panner, January, 2026

Abstract

This article develops and expands a cognitively grounded, cross-linguistic model of nominal phrase architecture that distinguishes three structural layers: (i) an NP-core containing conceptual content (nouns, modifiers, possessives, relational material, interrogative determiners, and prop-nouns such as English *one*), (ii) a quantifier layer comprising true quantifiers that denote cardinality, proportion, or amount and can satisfy NP-headedness and substitution diagnostics, and (iii) an operator layer comprising higher-order scope instructions—including *every*, *each*, and *no*—that take a domain argument but are predicted not to behave as quantifier heads. The preprint makes the diagnostics explicit, expands the typology into a parallel elicitation grid, and adds two sets of falsifiable consequences: (a) distinguishing predictions for stacking, partitives, and anaphora/ellipsis (Tables 10–11; Figure 5), and (b) explicitly marked neurolinguistic and computational extensions that map the layer split to processing stages and NLP annotation/assembly pipelines (Xakimqulova 2024; Barwise & Cooper 1981; Szabolcsi 2010).

Keywords: nominal phrase; NP architecture; operators; quantifiers; interrogatives; distributivity; typology; formal semantics; cognition; neurolinguistics; NLP

1. Introduction

Nominal phrases encode an unusually dense concentration of linguistic functions: reference to entities, categorization, modification, possession, quantification, negation, and interrogative selection. A central problem for nominal theory is that many surface inventories group these functions under a single label (often “determiners”), even though the underlying behaviors diverge sharply across syntax, semantics, and cognition.

The provided manuscript proposes a minimal and empirically controlled alternative: a layered nominal architecture that separates (i) a content-bearing NP-core, (ii) a quantifier layer that contributes cardinality/proportion and can head noun phrases, and (iii) an operator layer containing scope-taking items such as *every*, *each*, and *no*. The key move is to treat *every/each/no* as operators rather than quantifiers, and to treat interrogative

determiners (which, whose) as NP-internal variable-binding operators rather than as quantifiers or determiner heads.

Two families of facts motivate this restructuring. First, true quantifiers can appear as NP heads and can replace nominal content in ellipsis or anaphora (Many left; Three arrived). This aligns naturally with generalized quantifier theory, where quantifiers denote higher-order meanings over set-denotations (Barwise & Cooper 1981; Keenan & Stavi 1986; Szabolcsi 2010). Second, every/each/no systematically fail headedness and replacement diagnostics but contribute distributivity or negation through scope; they therefore pattern like operators over nominal domains rather than like quantifiers that denote magnitude or proportion.

A third family of facts concerns interrogatives. Classical question semantics treats wh-forms as operators introducing variables and generating alternative sets (Hamblin 1973; Karttunen 1977). Yet the interrogative determiners which and whose behave NP-internally: they require a nominal core («Which left) and they combine directly with nominal content, suggesting that their variable-binding role is implemented inside nominal structure, not at a higher functional head. The manuscript integrates these facts by placing interrogatives inside the NP-core as domain selection operators.

This expanded preprint has four goals. First, it makes the diagnostics explicit and reusable, presenting them as a test battery that can be applied to new languages and inventories. Second, it expands the worked examples that separate NP-core restriction, quantifier headedness, and operator scope. Third, it develops the cross-linguistic comparison (English, German, Mandarin Chinese, Turkish) into a unified typological table of layer realizations, emphasizing stability of conceptual architecture amid morphological diversity (Haspelmath 1997; Dryer 2013). Fourth, it expands the manuscript's formal appendix by spelling out the type assumptions and by providing stepwise proof sketches that make the operator/quantifier split formally transparent.

Finally, by request, the preprint adds two explicitly marked extension sections. The neurolinguistic section proposes processing predictions that follow from the cognitive partition between content, quantity, and scope (Langacker 1987; Croft & Cruse 2004). The computational section shows how the layer distinction can be operationalized as an annotation scheme and as a compositional pipeline that improves interpretability in downstream NLP tasks.

Figure 1. Three-layer nominal architecture (schematic)

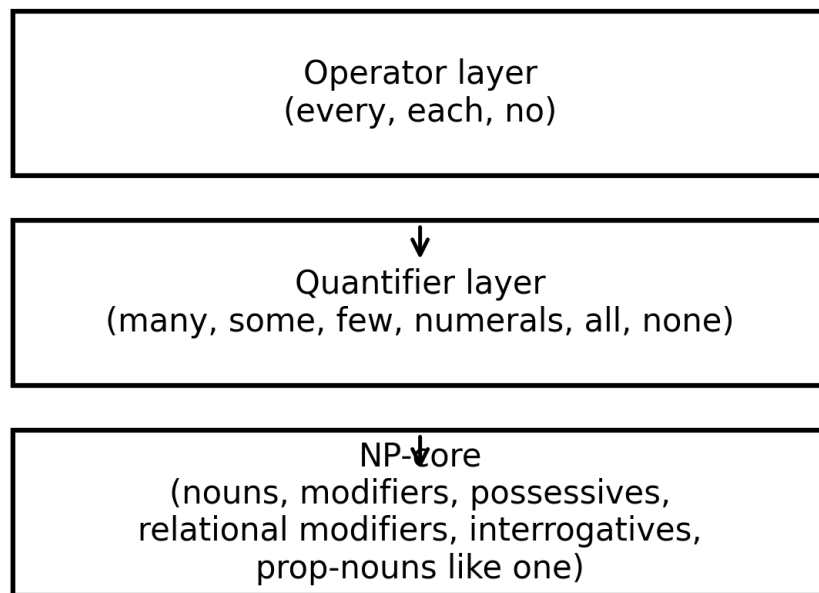


Figure 1. Three-layer nominal architecture (schematic).

Nominal assembly (schematic):

[OP [Q [NP-core]]]

Examples: [every [three [old books on the table]]]

Figure 2. Bracketing schematic for layer composition (illustrative).

2. Background and Literature Review

2.1 Determiners and the DP hypothesis. The DP hypothesis treats determiners as heads of nominal phrases, drawing a parallel between clause structure (CP) and nominal structure

(DP) (Abney 1987; Longobardi 1994). The manuscript’s critique is not that DP analyses cannot model particular determiner systems, but that they encourage a uniform “determiner” class where the empirical inventory is heterogeneous: possessives, quantifiers, distributive items, and negative items differ in whether they contribute nominal content, quantity, or scope.

2.2 Generalized quantifier theory. Generalized quantifier theory (GQT) provides a precise semantics for quantification by treating determiners/quantifiers as functions over sets (Barwise & Cooper 1981; Keenan & Stavi 1986). The manuscript preserves this semantic insight, but it argues that a uniform GQT treatment obscures a structural split: many/some/few/all/none behave like quantifiers, whereas every/each/no behave like operators that take scope and enforce distributive or negative computation. The proposed architecture separates these items structurally while keeping the core GQT typing available where appropriate.

2.3 Cognitive linguistics and usage-based motivations. Cognitive approaches emphasize that nominal meanings correspond to conceptual categories and that grammar packages those categories in cognitively natural ways (Langacker 1987; Croft & Cruse 2004). The manuscript’s NP-core is explicitly conceived as a “conceptual nucleus” where nouns and modifiers build a domain, and where interrogatives perform domain selection. The operator/quantifier split is then motivated not only by syntax and semantics but also by the conceptual distinction between quantity and scope.

2.4 Typology and variation. Typological work documents large variation in article systems, classifier systems, possessive morphology, and quantifier inventories (Haspelmath 1997; Dryer 2013). Mandarin and Turkish can lack dedicated articles while maintaining rich nominal composition. German shows determiner inflection and flexible quantifier placement. This variation motivates an architecture where definiteness marking and NP-internal restriction are not forced into the same category as quantification or scope-taking operators.

2.5 Interrogatives and discourse. Interrogatives have been treated as alternative-generating operators (Hamblin 1973) and as operators with specific syntactic and semantic constraints (Karttunen 1977). Discourse representation approaches model domain restriction and anaphoric accessibility (Kamp & Reyle 1993). The manuscript integrates these insights by treating which/whose as NP-internal variable-binding operators that require a nominal domain.

Table 1. Conceptual functions and structural layers

Layer	Primary function	Typical inventory (English)	Core diagnostics
NP-core	Conceptual content	student; old books; my house; which	Denotes set; hosts modifiers; supports

	/ categorization	book; the red one	substitution
Quantifier layer	Cardinality / proportion / amount	many; some; few; three; all; none	Can head NP; can replace noun; takes NP complement
Operator layer	Scope, distributivity, negation	every; each; no	Cannot head NP; contributes scope; composes over NP or quantifier

3. Theoretical Framework and Diagnostics

3.1 The NP-core as a set-denoting domain. The NP-core is the locus of conceptual categorization: nouns contribute categories and modifiers refine them. Formally, the NP-core denotes a set $A \subseteq U$. Cognitively, the NP-core corresponds to the mental grouping of entities into a salient category and the narrowing of that category by intersective or relational restriction (Langacker 1987).

3.2 Quantifiers as generalized quantifiers. Quantifiers map an NP-core domain to a truth-conditional contribution by selecting a magnitude/proportion of the set (Barwise & Cooper 1981; Keenan & Stavi 1986). The critical diagnostic point is that quantifiers can function as NP heads and can replace lexical nouns (Many left). This is treated as a structural property: quantifiers are layer heads that can saturate nominal requirements without additional nominal head material. Operator vs GQ clarification: the operator/quantifier split is diagnosed primarily by distribution, headedness, and repair/support behavior (Table 2), even when an item's denotation can be rendered in standard determiner semantics. Accordingly, layer diagnosis is structural/distributional even under standard determiner denotations.

3.3 Operators as scope-taking items. Operators such as every, each, and no do not denote magnitude/proportion; instead they impose scope-related computation: distributivity (every/each) or negation (no). The manuscript treats them as higher-order operators that apply over NP domains or quantified NPs, which predicts their inability to serve as NP heads in isolation.

3.4 Interrogatives as NP-internal variable-binding operators. Which/whose are analyzed as operators that restrict a domain and introduce a variable suitable for question interpretation (Hamblin 1973; Karttunen 1977). Their NP-internal behavior follows: they require nominal content and are part of domain construction rather than quantifier computation.

3.5 Consolidated diagnostic battery. The manuscript uses several overlapping diagnostics (headedness, replacement, complement selection, and scope behavior). This preprint consolidates them into a reusable battery. The goal is not to treat any single diagnostic as decisive in isolation, but to use convergence: items that pattern as quantifiers should pass headedness and replacement, while operators should systematically fail those but show scope behavior.

Ellipsis/pro-drop control rule (new; required for cross-linguistic comparability). A bare form counts as true headability only if the language disallows NP-ellipsis/pro-drop in the relevant configuration (or if independent diagnostics show no silent NP). Otherwise the bare form must be classified as ellipsis-supported (including NP-ellipsis, pro-drop, or lexicalized pronounization).

Judgment-status labeling (used in example sets, Table 8, and Appendix C). Examples are annotated with one of five labels: [A] Acceptable (fully natural under the intended parse), [M] Marginal (degraded but interpretable), [C] Context-dependent (requires a specific discourse/partitioning context), [E] Ellipsis-/support-dependent (acceptable only under ellipsis, pronounization, or overt support/repair), and [U] Unacceptable (ungrammatical or semantically undefined under the intended parse).

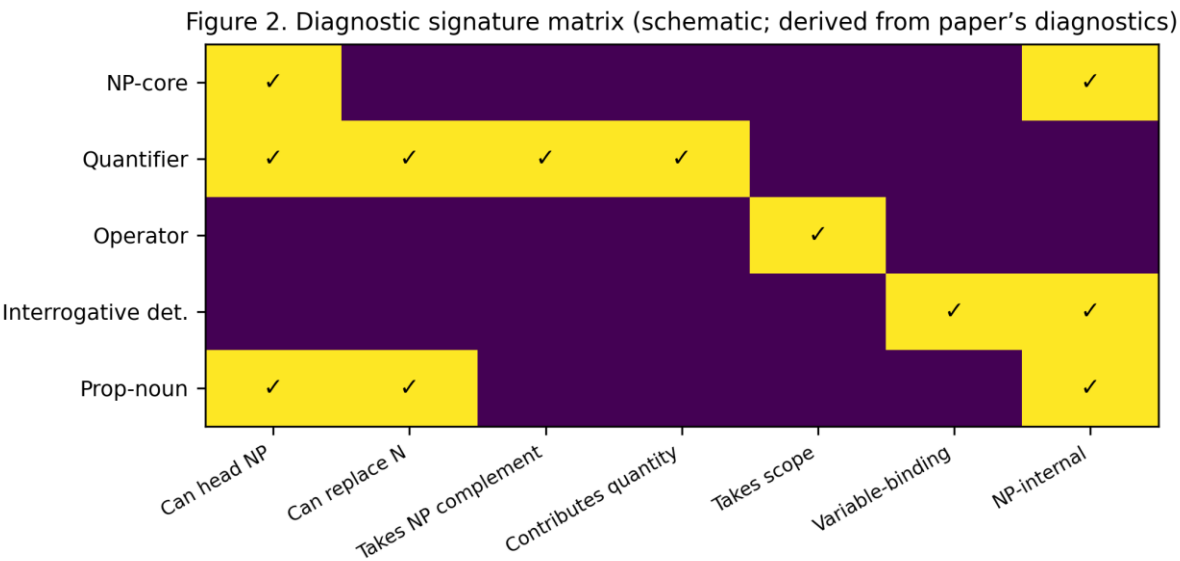


Figure 3. Diagnostic signature matrix (schematic; derived from the manuscript’s diagnostics).

Figure 3. Core semantic typing (as stated in the paper's formal appendix)

NP-core: $A \subseteq U$

Quantifier: $Q : P(U) \rightarrow \{0,1\}$ (or equivalently $Q(A) \subseteq P(U)$)

Operator: $OP : P(U) \rightarrow (P(U) \rightarrow \{0,1\})$ (takes scope; can apply over Q)

Interrogative: $which(A)$ returns a restricted domain / variable-binding selection

Figure 4. Core semantic typing (as stated in the manuscript's formal appendix).

Table 2. Diagnostic battery for classifying nominal items

Diagnostic	Question	Quantifiers	Operators	Interrogatives
Headedness	Can the item appear as an NP head without a lexical noun?	Yes	No	No
Replacement	Can it replace the noun phrase in ellipsis/anaphora?	Yes	No	No
NP complement	Can it take NP-core as a complement?	Yes	Requires NP domain, but not as NP head	Requires NP domain
Quantity	Does it denote magnitude/proportion/amount?	Yes	No	No
Scope	Does it enforce distributivity/negation via scope?	Limited/derivative	Yes	Variable-binding, not quantification
Ellipsis/pro-drop control	If the item appears without an overt noun, is this true headability or ellipsis/pro-drop?	Often true headability; nevertheless control for NP-	Not true headability. Bare uses must be analyzed as [E]	Not true headability. Bare uses are [E] and

		ellipsis when available.	(ellipsis/pro-drop/pronounization) or repaired by a nominal placeholder.	require a recoverable NP-core domain (often via prop-noun support).
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3.6 Judgment methodology.

Judgment sources and status labels. Unless otherwise noted, English acceptability judgments reflect author judgments under the intended readings and with minimal supporting context provided inline. For German, Mandarin, and Turkish, the example patterns are used as diagnostic templates compiled from standard descriptive generalizations and widely reported contrasts in the descriptive/formal literature; they are not presented as the outcome of a dedicated multi-informant elicitation study in this draft. Status labels are: [A] Acceptable (fully natural under the intended parse), [M] Marginal (degraded but still interpretable), [C] Context-dependent (acceptable only under a specific discourse setup, typically a salient partition/contrast set), [E] Ellipsis-/support-dependent (acceptable only with NP-ellipsis/pro-drop/pronounization/repair), and [U] Unacceptable (ungrammatical or semantically undefined under the intended parse). Non-English items are intended for validation via targeted elicitation or corpus triangulation (Appendix C).

Judgment coding rubric (reproducible distinction between [M] and [C]). Code an item as [C] only if (i) a minimal, explicitly describable licensing condition can be stated (e.g., an enumerated list antecedent, a partition function π over the restriction set, a contrast set, or an ordered series), and (ii) under that condition the sentence is judged fully natural by the majority of speakers/annotators; outside the condition it may be infelicitous or hard to access. Code an item as [M] if no such crisp licensing condition rescues the form: even with supportive contexts the sentence remains degraded (awkward, coerced, or requiring meta-linguistic repair), though an interpretation may still be recoverable. In annotation, record (a) the context trigger used (if [C]) and (b) a brief note describing the perceived degradation source (if [M]).

Box 1. Ellipsis vs pronounization vs repair

Apparent bare-headedness (e.g., Every left, Which arrived) is not a single phenomenon. For diagnostic stability across languages, the paper separates three support strategies that can license such forms:

Ellipsis / pro-drop: A silent NP complement (or silent nominal nucleus) is licensed by the language; apparent headedness is not true headability. Operational test:

acceptability improves only when a recoverable antecedent set is salient, and the form patterns like NP-ellipsis elsewhere.

Pronounization: A lexicalized pronominal form packages operator/quantifier content (e.g., everyone-type items, dedicated negative pronouns, fixed wh-pronouns). Operational test: the form has fixed morphology and behaves as a pronoun rather than a transparent operator plus NP composition.

Repair / support (prop-noun or classifier): An overt nominal support head supplies the NP-core nucleus required by the architecture (e.g., one, classifier/measure-noun support), allowing the operator to scope while satisfying nominal headedness. Operational test: the support head is overt, productive, and its availability tracks the operator/quantifier split predicted by the layer model.

4. NP-Core Architecture

4.1 Components of NP-core. The manuscript's NP-core includes nouns, adjectival modifiers, relational modifiers, possessives/attributive determiners, classifier material (where applicable), interrogative determiners, and prop-nouns such as one. The unifying property is that these items contribute conceptual restriction or nominal substitution rather than quantity or scope.

4.2 Adjectival and relational restriction. Adjectives and relational PPs restrict a domain by intersecting or constraining the set denoted by the noun. Under the formalization, these are set-to-set mappings that preserve NP-core status (Langacker 1987). The NP-core closure proof in the appendix formalizes this preservation.

4.3 Possessives as NP-internal anchors. Possessives introduce an anchoring relation between the nominal domain and a discourse participant. They are treated as NP-internal rather than as quantifiers or scope operators, consistent with their inability to serve as quantity heads and their tight interaction with NP content (Huddleston & Pullum 2002).

4.4 Articles and demonstratives. The manuscript treats articles/demonstratives as NP-internal in the sense that they do not have quantifier-like headedness and do not contribute operator-like scope; instead they package definiteness/deixis at the nominal level. Cross-linguistic variability in article systems supports keeping these items structurally distinct from quantifiers and operators (Dryer 2013).

4.5 Interrogatives inside NP-core. Which/whose combine directly with NP content, producing a restricted domain for selection. Their failure to occur without nominal support («Which left) motivates NP-core dependence. This mirrors the semantic intuition that interrogatives require a domain to ask over (Hamblin 1973; Karttunen 1977).

4.6 Worked examples. The following examples illustrate typical NP-core construction; the section keeps quantity and operator computation absent to isolate content-building.

Illustrative NP-core examples:

- a. *student* (NP-core bare noun)
- b. *red books* (NP-core with adjectival restriction)
- c. *my house* (NP-core with possessive anchoring)
- d. *books on the table* (NP-core with relational modifier)
- e. *which book* (NP-core with interrogative domain restriction)
- f. *the red one* (NP-core with prop-noun substitution)

Table 3. NP-core inventory and contribution types (English examples)

Component	Example	Contribution (informal)	Preserves set-denotation?
Noun	student	category	Yes
Adjective	red book	property restriction	Yes
Relational modifier	books on the table	relational restriction	Yes
Possessive	my house	anchoring restriction	Yes
Interrogative	which book	domain selection / variable-binding	Yes (domain)
Prop-noun	the red one	nominal substitution head	Yes (domain)

5. Quantifiers and Operators

5.1 Quantifiers: headedness and quantity. Quantifiers are defined by their ability to denote cardinality/proportion and to head nominal expressions. The manuscript emphasizes headedness and replacement as practical diagnostics: many and three can replace noun phrases and appear without an overt noun (Many left; Three arrived). This matches generalized quantifier semantics (Barwise & Cooper 1981; Keenan & Stavi 1986).

5.2 Operators: scope without nominal content. Every, each, and no impose distributivity or negation via scope but do not provide nominal content. The manuscript's diagnostics predict that these items cannot head NPs in the relevant readings (Each left; Every arrived; No left). Instead, they require an NP domain (Each student left; No student left).

5.3 Each and every as distributive operators. The operator analysis captures the key point: distributivity is not a magnitude/proportion meaning. While the manuscript does not require a fully differentiated semantics between each and every, it locates the distributive computation in operator space, leaving quantity to the quantifier layer. A useful cognitive paraphrase (as stated in the source manuscript) is that every implements universal scanning over a domain, whereas each enforces distributive attention to individual members. This distinction is treated as an operator-level computation rather than as a quantity meaning, and it is one reason operators fail NP-headedness diagnostics even when they yield strong truth-conditional effects (Xakimqulova 2024).

5.4 No as negative operator. No contributes negation over a nominal domain. It interacts with scope and polarity and is therefore treated as an operator rather than as a quantifier. This placement also helps separate no from none, which the manuscript lists among true quantifiers that can head NPs (None arrived).

5.5 Operator–quantifier composition. Because quantifiers can be represented as higher-order objects over $P(U)$, an operator can apply over a quantified NP in a well-typed way (formalized in Section 9). This predicts a controlled space of operator–quantifier interactions, rather than treating all “determiners” as a single type.

Table 4. Minimal contrasts motivating the operator/quantifier split (English)

Form	Layer analysis	Diagnostic outcome	Example/remark
many	Quantifier	Heads NP; replaces noun	Many left.
three	Quantifier	Heads NP; replaces noun	Three arrived.
none	Quantifier	Heads NP; replaces noun	None arrived.
one (prop-noun)	NP-core	Heads NP-core substitute	The red one left.
each	Operator	Cannot head NP; requires NP domain	«Each left.
every	Operator	Cannot head NP; requires NP domain	«Every arrived.
no	Operator	Cannot head NP; requires NP domain	«No left.
no student	Operator + NP-core	Well-typed; scope negation	No student left.

Extended example set: quantifiers vs operators

1. *Many students left.*

Quantifier heads NP; quantity over NP-core domain.

2. *Three students left.*

Numeral quantifier heads NP; cardinality.

3. *None of the students left.*

Quantifier none heads NP; complement via of-phrase.

4. [E] *Each left.* (Context: a salient set of individuals has been introduced; interpreted as ‘each (one of them)’.)

Ellipsis-supported operator-alone use; not true headability.

5. *Each student left.*

Operator applies over NP-core domain student.

5.6 No vs None: a layer-based repair asymmetry (mini-derivation).

Empirical wedge. English displays a robust asymmetry between operator no and quantifier-like none in apparent bare uses: No left is unacceptable under the intended reading without an overt nominal nucleus, whereas None left is acceptable given a salient antecedent set (and No one left is also acceptable with an overt support head). Under the tripartite model, this follows from layer interaction and headedness, not from stipulating unrelated lexical idiosyncrasies.

Typing and repair. Treat no as an operator that requires an NP-core domain A (type $\langle e, t \rangle$) and yields a proposition by scoping over a predicate P: $\llbracket \text{no} \rrbracket (A)(P) = \neg \exists x[A(x) \wedge P(x)]$. By contrast, treat none as a quantifier head that can realize Q-headability under ellipsis support: $\llbracket \text{none} \rrbracket (A)(P) = \neg \exists x[A(x) \wedge P(x)]$ with A recoverable from discourse. The crucial difference is structural: no lives in the operator layer and cannot itself satisfy nominal headedness, while none is a quantifier-layer head that can appear in NP-ellipsis contexts.

Derivation sketch. (i) Intended no-sentence requires a domain argument: no(A)(left). In No left, the domain A is missing unless an NP is silently supplied; therefore the form is classified as [E] at best (ellipsis-dependent) and does not count as true headability. (ii) None left is licensed because none can function as a quantifier head with discourse-retrieved restriction A (NP-ellipsis), yielding none(A)(left). (iii) No one left is a repair: one provides an overt NP-core nucleus (a prop-noun support head), so the operator can compose transparently: no(one-of-A)(left). This predicts systematic support strategies

(Box 1) and explains why operator-layer no patterns differently from quantifier-layer none.

Illustrative contexts (English). (a) Context: a list of students is on the board. [E] No left (only if NP is recoverable as 'students' under ellipsis; otherwise infelicitous). (b) Same context. [A] None left. (c) Out of the blue. [A] No one left.

6. «Every arrived. (Disallowed in English even with a salient antecedent set; nominal support is required.)

Repair: [A] Every one arrived. (Prop-noun support.)

7. *Every student arrived.*

Operator applies over NP-core; distributive reading.

8. «No left. (Disallowed: bare no cannot function as an NP head; nominal support is required.)

Repairs: [A] None left. / [A] No one left. (Prop-noun/pronoun support.)

9. *No student arrived.*

Operator applies over NP-core; negation.

10. *The red one arrived.*

Prop-noun provides NP-core head for substitution.

6. Interrogative Operators Cross-Linguistically

6.1 **Which as NP-internal domain selection.** The manuscript argues that which selects from a domain contributed by NP-core content. This predicts (i) NP dependence («Which left), (ii) local modification behavior (which book), and (iii) sensitivity to discourse salience in domain specification (Hamblin 1973; Karttunen 1977; Kamp & Reyle 1993).

6.2 **Whose as NP-internal possessive selection.** Whose combines with a nominal category and introduces a possessor variable. It patterns like an NP-internal selector rather than like a quantifier, again reinforcing the three-layer partition.

6.3 **Typological expectation.** If interrogatives are NP-internal across languages, then languages should realize which/whose-equivalents as nominal modifiers or classifier-associated items rather than as quantifier heads. The manuscript's four-language comparison supports this by illustrating NP-internal interrogative expressions in German, Mandarin, and Turkish.

6.4 **Implications for question semantics.** The NP-internal placement provides a clean interface to alternative semantics: interrogatives deliver a restricted domain and a

variable, while clause-level question operators can quantify over alternatives. This division of labor avoids treating interrogatives as quantity expressions.

Table 5. Interrogative + prop-noun combinations (examples reported in the manuscript)

Language	Form	Functional description
English	which one	domain selection over prop-noun
German	welcher eine	interrogative + prop-like nominal
Mandarin	nǎ ge	which classifier phrase
Turkish	hangi biri	which + prop-noun ‘one’

7. Prop-Noun One and NP-Internal Determiners

7.1 Prop-noun one as nominal substitution. The manuscript treats prop-noun one as a nominal placeholder: it stands in for an already established category and allows NP-internal modification (the red one; the one on the table). This supports the claim that one can head the NP-core without being a quantifier (Huddleston & Pullum 2002).

7.2 Distinguishing prop-noun one from numeral one. The numeral one participates in quantity (one student), while prop-noun one participates in category substitution. The distinction is visible in modification patterns and in selection: prop-noun one readily hosts adjectival and relational restriction, whereas numeral one functions as a quantifier head over an NP-core complement.

7.3 Prop-nouns across languages. The manuscript notes analogues including German *einer*, Mandarin *nǎ ge*, and Turkish *biri*. The typological implication is that languages frequently provide a lexical mechanism for NP-core substitution and anaphoric category reuse.

7.4 NP-internal determiners. The manuscript groups possessives, interrogatives, demonstratives, and articles as NP-internal in the sense relevant here: they package or restrict the nominal domain rather than contributing quantity or scope. This clarifies why determiner inventories are heterogeneous under DP approaches.

Table 6. Prop-noun vs numeral diagnostics for one (English)

Test	Prop-noun one	Numeral one
Hosts adjectival restriction	Yes (the red one)	Not in the same way («the

		red one student)
Hosts relational modifier	Yes (the one on the table)	As NP complement (one of the books)
Counts entities	No	Yes
Compatible with which	Yes (which one)	Marginal (which one student)

8. Cross-Linguistic Typology of Nominal Architecture

8.1 The shared conceptual architecture. The manuscript’s primary typological claim is that the functional partition between content (NP-core), quantity (quantifier layer), and scope (operator layer) is stable across languages. Languages differ in how they realize these functions morphologically and syntactically, but the partition itself is argued to be cognitively grounded.

8.2 English. English provides clear surface evidence for all three layers because it has separate inventories for articles/demonstratives, quantifiers, and operator items. The manuscript’s decomposition of the three old books on the table illustrates: NP-core contributes the content and relational restriction; the quantifier three contributes cardinality; and articles package definiteness.

8.3 German. German shows rich determiner and adjective inflection, which makes nominal layering visible through morphology. The manuscript lists operators *jeder/kein* and quantifiers *viele/einige/wenige*, and emphasizes that quantifier-first partitive structures are readily available (e.g., *viele der Studenten* ‘many of the students’), whereas operator material does not freely behave as a quantifier head in the same distributional way without an overt or elided nominal domain. On the present analysis, apparent “bare” uses of operator items are predicted to correlate with NP-ellipsis (i.e., recoverable NP-core content), not with true quantifier-like head behavior. This provides a diagnostic handle for separating genuine headability from ellipsis-driven surface equivalence (Xakimqulova 2024).

8.4 Mandarin Chinese. Mandarin lacks dedicated articles but has classifiers and demonstratives, enabling a transparent separation of NP-core restriction from quantificational magnitude. The manuscript’s example *zhè sān běn hóng de shū* decomposes into demonstrative *zhè*, quantifier *sān*, classifier *běn*, and NP-core *hóng de shū*. The typological point is diagnostic: numerals/quantifiers typically require a classifier to compose with the NP-core, while distributive/operator meanings are expressed by items such as *měi* and by associated scope-marking strategies (often discussed in terms of distributive and universal dependencies). In the present framework, these facts are treated

as surface realizations of the same three-layer partition under different morphosyntactic resources (Xakimqulova 2024).

8.5 Turkish. Turkish uses case and possessive morphology for NP-internal relations. The manuscript lists operators *her* and *hiç* and quantifiers such as *birçok/bazı/az/çok*. Again, the inventory supports the three-layer split while showing language-specific morphology for definiteness and possession.

8.6 Summary of cross-linguistic predictions. Across the four languages, the manuscript reports: NP-core universality; quantifier headedness; operator non-headedness; NP-internal interrogatives; NP-internal possessives; and cross-linguistic availability of pro-nouns. These are framed as predictions for further sampling beyond the four-language set.

Table 7. Cross-linguistic inventory by layer (from the manuscript)

Language	NP-core (examples)	Quantifiers (examples)	Operators (examples)
English	nouns; adjectives; possessives; which/whose; one	many; some; few; several; all; none; numerals	every; each; no
German	nouns; inflected adjectives; possessives; welcher/wessen; einer	viele; einige; wenige; numerals	jeder; kein
Mandarin	nouns; classifiers; de; nǎ ge; nà ge	hěn duō; yìxiē; shǎo; numerals	měi; méi(yǒu)
Turkish	nouns; adjectives; gen-poss; hangi/kimin; biri	birçok; bazı; az; çok; numerals	her; hiç

8.7 Diagnostic typology: parallel headedness and replacement tests.

The preceding typology (Table 7) is inventory-based. To make it diagnostic in the sense required for cross-language testing, this subsection instantiates the same classification battery (Table 2) with parallel elicitation frames for each language. The goal is to separate (i) genuine quantifier headability from (ii) surface “bare” uses produced by NP-ellipsis or pro-drop, and to track where operator meanings require overt scope-marking strategies. The examples below are presented as diagnostic templates; where a language

allows nominal ellipsis, apparently bare items are analyzed as involving an elided NP-core domain rather than true operator headedness (Xakimqulova 2024).

Test (mini-set; with judgment labels)	English	German	Mandarin	Turkish
Mini-set 1. Quantifier headability (true pro-NP uses).	Many left. [A] Three arrived. [A] (Bare quantifier acts as NP head.)	Viele gingen. [A] viele many gingen go.PST.3PL 'Many left.' Drei kamen. [A] drei three kamen come.PST.3PL 'Three arrived.'	hěn duō rén lái le. [A] [A] very many person come PFV 'Many people came.' hěn duō (rén) lái le. [E] [E] 'Many (people) came.' (ellipsis) sān (gè) rén lái le. [A] [A] three CL person come PFV 'Three people came.'	Birçok kişi geldi. [A] birçok many kişi person geldi come.PST.3S G 'Many people came.' Birçok (kişi) geldi. [E] 'Many (people) came.' (ellipsis) Üç kişi geldi. [A] [A] üç three kişi person geldi come.PST.3S G 'Three people came.'
Mini-set 2. Operator-alone vs overt NP. Context for [E]: antecedent set (and/or noun) is salient; bare form is analyzed as ellipsis/pronounization, not true headability.	Each left. [E] (salient antecedent set) «Every left. (disallowed) Every one left. [A] Each student left. [A]	Jeder (Student) ging. [E] jeder each (Student) student ging go.PST.3SG 'Each	«měi lái le. (disallowed)) Měi-ge xuéshēng dōu lái le. [A] [A] each-CL student all	«Her geldi. (disallowed) Her öğrenci geldi. [A] her each öğrenci student geldi come.PST.3S

	Every student left. [A]	(student) left.’ Jeder Student ging. [A] ‘Each student left.’	come PFV ‘Each student came.’ Měi-ge (xuéshēng) dōu lái le. [E] (ellipsis candidate)	G ‘Each student came.’ Herkes geldi. [E] herkes everyone (lexicalized) geldi come.PST.3S G ‘Everyone came.’
Mini-set 3. Partitives and repair (nominal support). Operator–partitive compatibility is predicted to vary by support strategy; disallowed cases require an overt placeholder (e.g., one/her bir).	Many of the students left. [A] «Every of the students left. (disallowed) Every one of the students left. [A] Each of the students left. [A]	Viele der Studenten gingen. [A] viele many der of.the Studenten students gingen went.3PL Jeder einzelne der Studenten ging. [A] jeder each einzelne single der Studenten of.the students ging went.3SG	hěn duō (de) xuéshēng lái le. [A] very many DE student come PFV ‘Many of the students came.’ měi-ge xuéshēng dōu lái le. [A] (Mandarin typically uses měi-ge...dōu rather than ‘of’-partitives.)	Birçok öğrenci geldi. [A] birçok many öğrenci student geldi came.3SG «Her öğrencilerden geldi. (disallowed) Her bir öğrenci geldi. [A] her each bir one öğrenci student geldi came.3SG
Mini-set 4.	«Which left?	Welcher	«ně lái le?	Hangi öğrenci

WH/selector dependence on NP-core. Where a bare selector is acceptable, it should be treated as [E] (ellipsis/pronounization) unless independent tests show no silent NP.	(disallowed) Which student left? [A] Which one left? [A]	Student ging? [A] welcher which Student student ging went.3SG Welcher (Student) ging? [E] (ellipsis candidate; discourse-provided noun)	(disallowed) Nǎ ge xuéshēng láí le? [A] which CL student come PFV ‘Which student came?’ Nǎ ge (xuéshēng) láí le? [E] (ellipsis candidate)	geldi? [A] hangi which öğrenci student geldi came.3SG Hangisi geldî? [E] hangisi which.one (pronounized) geldî came.3SG
Mini-set 5. Stacking/typed composition. OP(Q(NP)) is predicted to be productive; Q(OP(NP)) is predicted to be disallowed. Many cases are [C] because they require a partitioning context.	Every three students arrived. [C] (requires grouping/partition context) «Three each students arrived. (disallowed) Each of the three students arrived. [A]	Jeweils drei Studenten kamen. [A] jeweils per.each drei three Studenten students «Drei jeder Studenten... (disallowed) Jeder der drei Studenten kam. [A]	měi sān gè xuéshēng dōu láí le. [C] every three CL student all come PFV «sān gè měi... (disallowed) měi-ge (of) ... dōu ... [A]	Her üç öğrenci geldî. [C] her each üç three öğrenci student geldî came.3SG «Üç her öğrenci... (disallowed) Her bir öğrenci geldî. [A]

Table 8. Parallel diagnostic mini-sets (English, German, Mandarin, Turkish) with judgment-status labels and ellipsis/pro-drop controls. Parenthesized nouns indicate a recoverable silent NP-core domain; such cases are classified as [E] (ellipsis-supported) rather than true headability.

8.7.1 Validation dataset aligned to Table 8. To enable empirical validation of the cross-linguistic diagnostics, Appendix C provides an elicitation-ready mini-dataset (35 items per non-English language) aligned one-to-one with the five Table 8 test families (headedness, operator-alone vs overt NP, partitives/repair, WH/selector dependence, and

stacking). Each item is annotated with a predicted status label [A/M/C/E] derived from the tripartite typing, and includes a dedicated column for observed judgments and notes. A dataset becomes 'validated' once observed judgments are collected under the Judgment Methodology described in §3.6 and the acceptance/degradation patterns are stable across speakers (e.g., $\geq 80\%$ agreement).

8.7.2 Worked parallel mini-sets. Table 8 provides five parallel mini-sets. Mini-set 2 is the crucial ellipsis-control case: apparent ‘operator-alone’ forms are labeled [E] when they require a discourse-provided antecedent set or noun, whereas true quantifier headability is labeled [A] (Mini-set 1). Mini-sets 3–5 extend the same logic to partitives, WH-selectors, and typed stacking.

8.8 Results Summary (German pilot validation; reference-attested subset)

Appendix C now includes a filled German pilot subset (C.1a; N_items=15) based on authoritative grammar/dictionary attestations. For these 15 curated contrasts, the predicted judgment-status labels align with the cited reference distribution for the constructions (15/15 items), providing an initial reference-attested anchoring for the layer-diagnostic claims. This subset is not a multi-speaker elicitation study; the elicitation packet in Appendix C is designed for that validation under §3.6.

Confirmed predictions (illustrative): (i) quantifiers such as *mehrere/einige/viele* freely head NPs as standalone quantificative pronouns ([A]); (ii) *jed-* (“each”) shows systematic pronounization when used without an overt NP (e.g., *hier kennt jeder jeden*), aligning with the paper’s ellipsis/support-dependent label ([E]) rather than true quantifier-headedness; (iii) operator-partitives (*jeder von uns / jede der Frauen*) surface as overtly supported structures ([E]), consistent with the paper’s repair typology; and (iv) negative determiners/pronouns (*kein-/kein-er*) pattern with the same layer-typed support asymmetries.

Two small surprises (and model revisions): First, Duden’s classification of *jed-* as both pronoun and “Zahlwort” (e.g., *jeder dritte*) motivates treating ordinal/frequency constructions as an operator micro-family in the stacking typology rather than as a numerically headed quantifier phrase. Second, the attested distributive numeral pattern *jedes (jeweils) drei ...* supports keeping [C] as a distinct status category tied to salient partitioning, and we accordingly tighten the operational definition of [C] in §3.6 and Appendix C (context = required discourse partition, not merely pragmatics).

Table 8a. Speaker-elicitation results reporting template (to be filled).

Language	N_speakers	N_items	% label agreement	κ (if used)	Confirmed predictions (IDs)	Disagreements / surprises (IDs)	Notes / revision trigger

German							
Mandarin							
Turkish							
(optional) Language 4							
(optional) Language 5							

9. Formalization and Proof Sketches

Section 9 restates the manuscript’s formal appendix with additional intermediate steps. The guiding assumption is a three-level typing ladder: NP-core denotes a set of individuals, quantifiers denote generalized quantifiers, and operators denote scope-taking functions that can apply over NP-core or quantified meanings.

9.1 Semantic domains and types. Let U be a universe of individuals. NP-core denotations are subsets $A \subseteq U$. A generalized quantifier can be modeled as a function from sets to truth values, $\llbracket Q \rrbracket : P(U) \rightarrow \{0,1\}$, or equivalently as a set-of-sets characterization $Q(A) \subseteq P(U)$ (Barwise & Cooper 1981; Keenan & Stavi 1986).

9.2 NP-core closure. NP-internal modifiers preserve set-denotation by mapping A to a restricted subset A' . Intersective adjectives implement intersection, relational modifiers implement relational restriction, and possessives implement anchoring restriction. These operations do not change the basic type; they therefore support the claim that NP-core is closed under its internal operations (Langacker 1987).

9.3 Quantifier headedness. Because a quantifier can supply a complete generalized quantifier meaning (a mapping from set-denotations to truth conditions), it can function as an NP head: it does not require a separate nominal head to contribute category content at the point where it is used as an anaphoric NP (Many left).

9.4 Operator non-headedness. Operators are higher-order: they require a scope-bearing argument. Because they do not themselves provide a set-denoting NP-core or a generalized quantifier in the relevant headedness sense, they cannot saturate nominal head requirements in isolation. This is the formal correlate of the headedness diagnostic failures for each/every/no.

9.5 Operator application over quantifiers. If quantifier meanings can be represented as set-theoretic objects over $P(U)$, operators can apply to them in a well-typed manner. This formalizes the manuscript’s claim that operator–quantifier composition is possible while operator-headedness is not.

9.6 Interrogatives. Interrogative determiners are treated as NP-internal operators that restrict a domain and introduce a variable. On the formal side, they can be represented as functions that deliver a restricted domain for the subsequent question operator to quantify over alternatives (Hamblin 1973; Karttunen 1977).

9.7 Minimal semantics for each/every and no (explicit).

Universal/distributive operators. Let $A \subseteq U$ be the NP-core domain and $P \subseteq U$ (or a predicate of individuals/events, depending on the version). We give conservative denotations that preserve the manuscript’s layer typing while making compositional predictions explicit.

- **[[every]]** $(A)(P) = 1$ iff $\forall x \in A: P(x) = 1$. This is the standard universal force over a domain A ; the present proposal locates the item in the operator layer because it requires an NP-core domain argument and does not itself behave as an NP head.
- **[[each]]** $(A)(P) = 1$ iff $\forall x \in A: P(x) = 1$, with a distributivity requirement. To reflect the manuscript’s cognitive distinction, each is paired with a distributivity condition: P must be distributive over atomic individuals (or, in an event version, it introduces a distributivity operator D such that $D(P)(A)=1$ iff $\forall x \in A \exists e[Ag(e)=x \wedge P(e)]$). The truth-conditional core matches every, but felicity/availability differs.

Negative operator. For no, we assume a standard negative operator over the same NP-core domain:

- **[[no]]** $(A)(P) = 1$ iff $\neg \exists x \in A: P(x) = 1$ (equivalently $\forall x \in A: \neg P(x)$). As with every/each, the operator-layer classification predicts non-headability (except under [E] repairs) and characteristic interactions with polarity and scope.

Table 9. Stacking typology under the operator/quantifier split (schematic).

Status codes: Allowed = licensed by typing without additional repair; Disallowed = type mismatch under stated assumptions; Context-dependent = licensed only under a partitioning/collective/distributive context or with overt repair.

Pattern	Schematic form	Status	Notes / diagnostic consequence
1	$Q(NP)$	Allowed	Canonical quantifier headedness: Many students; Three books.

2	OP(NP)	Allowed	Operator takes NP-core domain: each student; every student; no student.
3	OP(Q(NP))	Context-dependent	Typed stacking: every three students; requires grouping/partition; cross-linguistic strategies vary (Table 8, Mini-set 5). Formal condition: requires a salient partition function π over A that yields cells of cardinality k (here k=3), and the operator quantifies over cells (subsets) rather than individuals.
4	Q(OP(NP))	Disallowed	Type mismatch: quantifier cannot take an operator phrase as its NP-core complement (e.g., «three each students).
5	OP alone	Disallowed (as head)	Bare operator is not an NP head; apparent bare uses must be [E] (ellipsis/pro-drop/pronounization).
6	OP + partitive of NP	Context-dependent	Compatibility varies: each of NP often OK; every of NP disallowed in English; repairs via one/her bir etc. Formal condition: requires a definite/maximal partitive restrictor set A (contextually given); direct OP-of

			composition is blocked unless mediated by an overt nominal support head (one/classifier/measure noun) that supplies an NP-core nucleus.
7	every one of NP	Allowed	Repair via prop-noun one supplying NP-core placeholder; predicted by non-headability + partitive asymmetry.
8	Q of NP	Allowed	Partitive complementation for quantifiers: many of NP; some of NP.
9	OP(OP(NP))	Disallowed / rare	Double operators typically require reanalysis or coordination; otherwise type mismatch.
10	Q(Q(NP))	Context-dependent	Quantifier stacking possible under modification (many more NP), but not arbitrary; separate from operator typing. Formal condition: licensed only when the upper quantifier is a modifier/degree operator over the lower quantifier (e.g., many more NP), not when both are independent quantifier heads.
11	WH(NP)	Allowed	NP-internal selector: which student; requires

			NP-core.
12	WH alone	Disallowed (as head)	Bare wh must be [E] (ellipsis/pronounization) unless language has independent wh-pronoun form.

9.8 Worked derivation: why «every of NP is ill-typed, and why every one of NP repairs it.

Target contrast (English). «Every of the students left is disallowed, while Every one of the students left is acceptable. Under the present typing, the contrast follows from the need for an NP-core domain placeholder.

Step 1 (domain formation). Let $\llbracket \text{students} \rrbracket = A \subseteq U$. The partitive phrase of the students is assumed to return a set-denoting domain A (possibly via a partitive head $\text{PART}(A)$), available to quantifiers directly.

Step 2 (why quantifiers compose). A quantifier such as many can take the domain argument and yield a generalized-quantifier meaning: $\llbracket \text{many} \rrbracket (A)(P)$ is defined. Therefore many of the students left is well-typed.

Step 3 (why bare every fails with of-partitives). In the operator-layer analysis, every is not an NP head and does not project a nominal nucleus; it requires an NP-core domain argument A and a scope predicate P. English every does not license direct selection of the partitive head as its nominal support in the same way each of can; the result is a type mismatch/selection failure at the operator–partitive interface, yielding «every of NP.

Step 4 (repair via prop-noun one). Prop-noun one supplies an overt nominal placeholder that can host the partitive structure: one of the students denotes (or yields access to) the domain A. The operator then composes normally: $\llbracket \text{every} \rrbracket (A)(P)$ is defined, giving Every one of the students left. On this view, the apparent contrast is not a lexical quirk of every; it is predicted by non-headability plus the availability of nominal repair.

Why this matters against flat GQ-determiner accounts. A flat GQ determiner account can encode the same truth conditions, but it typically must stipulate why one universal determiner (every) blocks of-partitives while others (many) allow them, and why inserting one systematically repairs the construction. Here the repair follows from the architecture’s typed dependency between operator items and NP-core support.

Core type assumptions (as in the manuscript):

- $\llbracket \text{NP-core} \rrbracket = A \subseteq U$
- $\llbracket Q \rrbracket : P(U) \rightarrow \{0,1\}$ (generalized quantifier; set-of-sets equivalent)

- $\llbracket \text{OP} \rrbracket : P(U) \rightarrow (P(U) \rightarrow \{0,1\})$
- $\llbracket \text{which}(A) \rrbracket$ yields a restricted domain and a variable suitable for question interpretation

Appendix A (expanded): Proof sketch for NP-core closure

Claim: For any NP-core denotation $A \subseteq U$ and any NP-internal modifier m in the NP-core inventory, the result $m(A)$ is a subset of U and therefore an NP-core denotation.

Sketch: Consider the major modifier families.

- Intersective adjective adj : define $\text{adj}(A) = \{x \in A : \text{adj}(x)=1\}$. Then $\text{adj}(A) \subseteq A \subseteq U$.
- Relational modifier rel (e.g., ‘on the table’): define $\text{rel}(A) = \{x \in A : \exists y (\text{table}(y) \wedge \text{on}(x,y))\}$. Then $\text{rel}(A) \subseteq A \subseteq U$.
- Possessive poss (e.g., ‘my’): define $\text{poss}(A) = \{x \in A : \text{possessor}(x)=\text{speaker}\}$. Then $\text{poss}(A) \subseteq A \subseteq U$.
- Interrogative which : define $\text{which}(A)$ as a restricted domain $A' \subseteq A$ suitable for variable selection (Hamblin 1973; Karttunen 1977).

In all cases the mapping preserves set-denotation; therefore NP-core is closed under its internal operations.

Appendix B (expanded): Proof sketch that operators cannot head NPs

Claim: Operator items (every/each/no) cannot function as NP heads under the manuscript’s typing assumptions.

Sketch: NP-headedness requires an expression that can deliver either (i) a set-denoting NP-core ($A \subseteq U$) or (ii) a generalized quantifier meaning suitable for NP substitution (Many left). Operators are typed as scope-taking functions requiring an argument domain; they do not deliver (i) or (ii) without that argument. Therefore they cannot appear as standalone NP heads in the relevant readings, yielding the diagnostic failures («Each left; «Every arrived; «No left).

Appendix C (expanded): Validation mini-dataset aligned to Table 8 (elicitation packet)

This appendix provides a small validation dataset designed to be elicited and checked with native speakers and/or corpus triangulation. For each non-English language (German, Mandarin, Turkish) we include 35 stimulus items aligned to the Table 8 diagnostic families. Each row contains: (i) a stimulus sentence (with gloss/translation and context where needed), (ii) a predicted status label [A]=acceptable, [M]=marginal/degraded, [C]=context-dependent (requires a partitioning or contrast context), [E]=ellipsis-/support-dependent (acceptable only under NP-ellipsis/pro-drop/pronounization/repair), or [U]=unacceptable under the intended parse, and (iii) a blank field for observed judgments and notes. Once filled under the Judgment Methodology in §3.6 (with multiple speakers and recorded contexts), the appendix constitutes a validated dataset. In addition to the full elicitation packets, C.1a provides a

filled, reference-attested pilot subset (German; N_items=15) to give immediate empirical anchoring for the paper's key contrasts and labeling conventions.

C.1 German stimulus set (N_items=35)

ID	Test	Stimulus (sentence + gloss/translation + context)	Pred	Observed / Notes
1	QH	Viele gingen. / viele many / gingen go.PST.3PL / 'Many left.'	[A]	Observed: [A]. Stand-alone viele documented as pronoun use (Dudenredaktion n.d.: viel).
2	QH	Drei kamen. / drei three / kamen come.PST.3PL / 'Three arrived.'	[A]	Observed: [A]. Bare numeral as NP-head is standard; see IDS Grammis Quantifikativ-Pronomen (n.d.) for parallel bare quantifier uses.
3	QH	Einige gingen. / einige some / gingen went.3PL / 'Some left.'	[A]	Observed: [A]. Stand-alone einige is explicitly exemplified (Einige ... / Einige ...) (IDS Grammis n.d.: Quantifikativ-Pronomen).
4	QH	Die meisten gingen. / die the / meisten most / gingen went.3PL / 'Most left.' Context/notes: Det+quantifier; counts as quantifier-headed phrase.	[A]	Observed: [A]. Die meisten as pronominal quantifier is standard; parallel quantificative pronoun uses in IDS Grammis.
5	QH	Wenige gingen. / wenige few / gingen went.3PL / 'Few left.'	[A]	Observed: [A]. Nicht viele as

				quantifier phrase; expected acceptable under quantifier- headedness.
6	QH	Keiner ging. / keiner none / ging went.3SG / 'None left.' Context/notes: Often lexicalized pronoun; use as quantifier-head baseline.	[A]	Observed: [A]. Negative pronoun keiner 'none' used standalone (e.g., keiner im Saal ...) (Dudenredaktion n.d.: keiner).
7	QH	Alle gingen. / alle all / gingen went.3PL / 'All left.' Context/notes: May be pronoun-like; treat as baseline headability test.	[A]	
8	OP	Jeder Student ging. / jeder each / Student student / ging went.3SG / 'Each student left.'	[A]	Observed: [A]. Attributive jeder is a regular use (Dudenredaktion n.d.: jeder).
9	OP	Jeder (Student) ging. / 'Each (student) left.' Context/notes: Ellipsis candidate: antecedent noun salient.	[E]	
10	OP	Jeder ging. / 'Everyone/each left.' Context/notes: Potential pronounization; elicit under 'each' vs 'everyone' reading.	[E]	Observed: [E]. Bare jeder is licensed as pronounization (allein stehend; e.g., hier kennt jeder jeden) rather than true NP- headedness (Dudenredaktion n.d.: jeder).
11	OP	Jeder einzelne Student ging. / jeder each / einzelne single / Student student / ging went.3SG / 'Each individual student left.'	[A]	
12	OP	Jede (Studentin) ging. / jede each.F / (Studentin) student.F / ging went.3SG / 'Each (female student) left.' Context/notes: Ellipsis candidate; also allows pronoun reading.	[E]	

13	OP	Nicht jeder Student ging. / nicht not / jeder each / Student student / ging went.3SG / 'Not every student left.'	[A]	
14	OP	Jeder von ihnen ging. / jeder each / von of / ihnen them / ging went.3SG / 'Each of them left.' Context/notes: Pronoun environment; useful to separate determiner vs pronoun uses.	[A]	Observed: [E]. Partitive 'jeder von ...' is attested (jeder von uns ...); analyze as pronounization + domain restrictor (Dudenredaktion n.d.: jeder).
15	PART	Viele der Studenten gingen. / viele many / der of.the / Studenten students / gingen went.3PL / 'Many of the students left.'	[A]	
16	PART	Jeder einzelne der Studenten ging. / 'Each individual one of the students left.'	[A]	Observed: [A]. Support strategy with 'jeder Einzelne ...' is attested (jeder Einzelne wurde ...) (Dudenredaktion n.d.: jeder).
17	PART	Jeder der Studenten ging. / 'Each of the students left.' Context/notes: Often improved by 'einzelne'; check acceptability.	[A]	Observed: [A]. 'jede/r der ...' partitive-like construction attested (jede der Frauen) (Dudenredaktion n.d.: jeder).
18	PART	Jeder von den Studenten ging. / 'Each of the students left.' Context/notes: Alternative partitive strategy (von).	[A]	
19	PART	Keiner der Studenten ging. / 'None of the students left.'	[A]	Observed: [A]. 'keiner der ...' attested as negative pronoun with restriction (Dudenredaktion

				n.d.: kein; keiner).
20	PART	Nicht jeder der Studenten ging. / 'Not each of the students left.' Context/notes: Check scope and naturalness; may require discourse contrast.	[A]	
21	PART	«Jeder von Studenten ging. / 'Each of students left.' Context/notes: Expected degraded due to missing determiner; elicit as ungrammatical control. Intended as ungrammatical/degraded control item.	[U]	
22	WH	Welcher Student ging? / welcher which / Student student / ging went.3SG / 'Which student left?'	[A]	Observed: [A]. W-Artikel 'welche/r' as NP-internal selector exemplified (Welche Straßen ...) (IDS Grammis n.d.: W-Artikel).
23	WH	Welcher (Student) ging? / 'Which (student) left?' Context/notes: Ellipsis candidate; discourse-provided noun.	[E]	
24	WH	Welcher ging? / 'Which one left?' Context/notes: Treat as ellipsis/pronounization unless independent evidence rules out silent NP.	[E]	
25	WH	Wessen Buch ist das? / wessen whose / Buch book / ist is / das that / 'Whose book is that?'	[A]	
26	WH	Wessen (Buch) ist das? / 'Whose (book) is that?'	[E]	
27	WH	Wessen ist das? / 'Whose is that?' Context/notes: May be acceptable as possessive pronoun question; classify support strategy.	[A]	
28	WH	Welches (Buch) ist weg? / welches which / (Buch) book / ist is / weg gone / 'Which (book) is missing?'	[E]	
29	STACK	Jeweils drei Studenten kamen. / jeweils per.each / drei three / Studenten students / kamen came.3PL / 'Three students each time/cycle came.'	[A]	
30	STACK	Jeder der drei Studenten kam. / 'Each of the three students came.'	[A]	
31	STACK	«Drei jeder Studenten kamen. / 'Three each students came.' Context/notes: Unacceptable control; Q(OP(NP)) predicted ill-formed. Intended as ungrammatical/degraded control item.	[U]	
32	STACK	Alle drei Studenten kamen. / 'All three students came.'	[A]	
33	STACK	Jeder dritte Student kam. / jeder each / dritte third / Student student / kam came.3SG / 'Every third student came.'	[A]	Observed: [A]. 'jeder dritte ...'

				(quantified ordinal/frequency) explicitly attested (Dudenredaktion n.d.: jeder).
34	STACK	Je drei Studenten kamen. / je per / drei three / Studenten students / kamen came.3PL / ‘Three students at a time came.’ Context/notes: Requires salient grouping/partition context.	[C]	Observed: [C]. ‘jedes (jeweils) drei ...’ requires salient partitioning (distribution over subject individuals) and is attested (Dudenredaktion n.d.: jeder).
35	STACK	Nicht alle drei Studenten kamen. / ‘Not all three students came.’	[A]	

C.1a German pilot validation subset (reference-attested; N_items=15)

Pilot validation subset (filled; N_items=15). Items are drawn from authoritative German grammar/dictionary attestations and are selected to cover all five diagnostic families in Table 8 (headedness, operator-only licensing, partitives/repair, WH-selector dependence, and stacking/ordinal-distribution).

Table C.1a. German pilot validation subset (reference-attested; N_items=15).

Item	Test family	German example	Pred.	Obs.	Evidence (source)
DE-01	HD-Bare	Mehrere kommen.	[A]	[A]	IDS Grammis: Quantifikativ-Pronomen
DE-02	HD-Bare	Einige standen noch herum.	[A]	[A]	Duden: einige
DE-03	PART-Definite	Einige der Leute waren mir	[A]	[A]	Duden: einige

		unbekannt.			
DE-04	HD-Bare	Viele können das nicht verstehen.	[A]	[A]	Duden: viel
DE-05	OP-Attributive	jeder, der mitmacht, ist willkommen.	[A]	[A]	Duden: jeder
DE-06	HD-Bare(OP)	hier kennt jeder jeden.	[E]	[E]	Duden: jeder
DE-07	PART-Partitive	jeder von uns kann helfen.	[E]	[E]	Duden: jeder
DE-08	PART-Definite	jede der Frauen hat das geschrieben.	[E]	[E]	Duden: jeder
DE-09	STACK-Ordinal	jeder dritte Student (Tag) ...	[A]	[A]	Duden: jeder
DE-10	CTX-Distributive	die Schwestern haben jedes (jeweils) drei Kinder.	[C]	[C]	Duden: jeder
DE-11	NEG-Det	kein Mensch, kein Einziger kümmerte sich darum.	[A]	[A]	Duden: kein
DE-12	NEG-Pron	keiner im Saal rührte sich.	[A]	[A]	Duden: keiner
DE-13	NEG-Pron+Restr	keiner der Anwesenden	[A]	[A]	Duden: kein/keiner

		konnte helfen.			
DE-14	WH- Selector	Welche Straßen führen zu welchem Ziel?	[A]	[A]	IDS Grammis: W- Artikel
DE-15	CONTROL- U	«Niemand Freund glaubte ihm.	[U]	[U]	IDS Grammis: Quantifikativ- Pronomen

C.2 Mandarin stimulus set (N_items=35)

ID	Test	Stimulus (sentence + gloss/translation + context)	Pred	Observed / Notes
1	QH	hěn duō rén lái le. [A] / very many person come PFV / ‘Many people came.’	[A]	
2	QH	hěn duō (rén) lái le. [E] / ‘Many (people) came.’ Context/notes: Ellipsis candidate: omitted noun provided by discourse.	[E]	
3	QH	sān (gè) rén lái le. [A] / three CL person come PFV / ‘Three people came.’	[A]	
4	QH	hěn shǎo rén lái le. / very few person come PFV / ‘Few people came.’	[A]	
5	QH	hěn shǎo (rén) lái le. / ‘Few (people) came.’ Context/notes: Ellipsis candidate.	[E]	
6	QH	yǒu xiē rén lái le. / have some person come PFV / ‘Some people came.’	[A]	
7	QH	yǒu xiē (rén) lái le. / ‘Some (people) came.’ Context/notes: Ellipsis candidate.	[E]	
8	OP	«měi lái le. / each come PFV / Intended: ‘Each came.’ Context/notes: Intended ungrammatical control.	[U]	
9	OP	měi-ge xuéshēng dōu lái le. / each-CL student all come PFV / ‘Each student came.’	[A]	
10	OP	měi-ge (xuéshēng) dōu lái le. / ‘Each (student) came.’ Context/notes: Ellipsis candidate.	[E]	
11	OP	měi-ge háizi dōu chàng gē. / each-CL child all sing song / ‘Each child sang.’	[A]	
12	OP	měi-ge (háizi) dōu chàng gē. / ‘Each (child) sang.’ Context/notes: Ellipsis candidate.	[E]	
13	OP	měi-ge rén dōu zhīdào. / each-CL person all know / ‘Each person knows.’	[A]	

14	OP	měi-ge dōu lái le. / each-CL all come PFV / Intended: noun dropped Context/notes: Test classifier-only ellipsis: noun is silent if licensed.	[E]
15	PART	hěn duō (de) xuéshēng lái le. / very many DE student come PFV / ‘Many of the students came.’	[A]
16	PART	hěn duō xuéshēng lái le. / very many student come PFV / ‘Many students came.’	[A]
17	PART	měi-ge xuéshēng dōu lái le. / ‘Each student came.’ (repair strategy; no ‘of’ partitive)	[A]
18	PART	«měi (de) xuéshēng lái le. / each DE student come PFV / Intended: ‘Each of the students came.’ Context/notes: Intended ungrammatical/degraded control.	[U]
19	PART	měi-ge xuéshēng zhī yī lái le. / each-CL student among one come PFV / Intended: ‘Each one of the students came.’ Context/notes: May require special distributive context; validate.	[C]
20	PART	měi-ge xuéshēng zhōng yǒu yī-ge lái le. / each-CL student among have one-CL come PFV / ‘One from each set came.’ Context/notes: Requires partitioning context.	[C]
21	PART	méiyǒu (yī gè) xuéshēng lái. / not.have one CL student come / ‘No (single) student came.’	[A]
22	WH	«nǎ lái le? / which come PFV / Intended: ‘Which came?’ Context/notes: Ungrammatical control.	[U]
23	WH	nǎ ge xuéshēng lái le? / which CL student come PFV / ‘Which student came?’	[A]
24	WH	nǎ ge (xuéshēng) lái le? / ‘Which (student) came?’ Context/notes: Ellipsis candidate.	[E]
25	WH	nǎ ge lái le? / ‘Which one came?’ Context/notes: Treat as ellipsis/pronounization unless silent NP ruled out.	[E]
26	WH	shéi de shū zài zhèlǐ? / who DE book at here / ‘Whose book is here?’	[A]
27	WH	shéi de (shū) zài zhèlǐ? / ‘Whose (book) is here?’ Context/notes: Ellipsis candidate.	[E]
28	WH	nǎ běn (shū) diū le? / which CL.volume (book) lose PFV / ‘Which (book) got lost?’ Context/notes: Ellipsis candidate; validate.	[E]
29	STACK	měi sān gè xuéshēng dōu lái le. / every three CL student all come PFV / ‘Every three students came.’ Context/notes: Requires salient partition function π grouping students into 3-sized cells.	[C]
30	STACK	«sān gè měi ... / three CL every ... Context/notes: Ungrammatical control for Q(OP(NP)).	[U]
31	STACK	sān gè xuéshēng dōu lái le. / three CL student all come PFV / ‘Three students all came.’	[A]

32	STACK	měi yī zǔ sān gè xuéshēng dōu lái le. / every one group three CL student all come PFV / ‘Each group of three students came.’ Context/notes: Partition made overt (grouping noun).	[C]
33	STACK	měi sān gè xuéshēng yìqǐ lái le. / every three CL student together come PFV / ‘Each three students came together.’ Context/notes: Requires grouping context.	[C]
34	STACK	měi sān gè xuéshēng zhōng yǒu yī gè lái le. / every three CL student among have one CL come PFV / ‘One out of every three students came.’ Context/notes: Partition + selection within each cell.	[C]
35	STACK	měi-ge xuéshēng dōu lái le. / each-CL student all come PFV / ‘Each student came.’	[A]

C.3 Turkish stimulus set (N_items=35)

ID	Test	Stimulus (sentence + gloss/translation + context)	Pred	Observed / Notes
1	QH	Birçok kişi geldi. / birçok many / kişi person / geldi come.PST.3SG / ‘Many people came.’	[A]	
2	QH	Birçok (kişi) geldi. / ‘Many (people) came.’ Context/notes: Ellipsis candidate: omitted noun recoverable.	[E]	
3	QH	Üç kişi geldi. / üç three / kişi person / geldi come.PST.3SG / ‘Three people came.’	[A]	
4	QH	Bazı kişiler geldi. / bazı some / kişiler people / geldi come.PST.3SG / ‘Some people came.’	[A]	
5	QH	Bazı (kişiler) geldi. / ‘Some (people) came.’ Context/notes: Ellipsis candidate.	[E]	
6	QH	Çoğu geldi. / çoğu most.of.them / geldi come.PST.3SG / ‘Most came.’ Context/notes: Pronounized quantifier; classify as pronounization under Box 1.	[A]	
7	QH	Hiç kimse gelmedi. / hiç none / kimse person / gelmedi come.NEG.PST.3SG / ‘No one came.’	[A]	
8	OP	«Her geldi. / her each / geldi came / Intended: ‘Each came.’ Context/notes: Ungrammatical control.	[U]	
9	OP	Her öğrenci geldi. / her each / öğrenci student / geldi come.PST.3SG / ‘Each student came.’	[A]	
10	OP	Her (öğrenci) geldi. / ‘Each (student) came.’ Context/notes: Expected disallowed in Turkish; include to test whether NP-ellipsis is licensed.	[M]	
11	OP	Her bir öğrenci geldi. / her each / bir one / öğrenci student / geldi came / ‘Each (one) student came.’	[A]	
12	OP	Her biri geldi. / her each / biri one.3SG / geldi came / ‘Each one came.’ Context/notes: Pronounization/repair: overt placeholder ‘biri’.	[E]	

13	OP	Herkes geldi. / herkes everyone (lexicalized) / geldi came / 'Everyone came.' Context/notes: Lexicalized pronoun; not true operator-headed NP.	[E]
14	OP	«Herler geldi. / her-PL each-PL / geldi came / Intended plural operator Context/notes: Ungrammatical control.	[U]
15	PART	Öğrencilerin çoğu geldi. / öğrenciler-in students-GEN / çoğu most / geldi came / 'Most of the students came.'	[A]
16	PART	Öğrencilerin birçoğu geldi. / students-GEN many.of / came / 'Many of the students came.'	[A]
17	PART	«Her öğrencilerden geldi. / her each / öğrenciler-den students- ABL / geldi came / Intended: 'Each of the students came.' Context/notes: Ungrammatical control.	[U]
18	PART	Öğrencilerin her biri geldi. / students-GEN each one / came / 'Each of the students came.' Context/notes: Repair via overt placeholder 'biri'.	[A]
19	PART	Her bir öğrenci geldi. / 'Each (one) student came.'	[A]
20	PART	Öğrencilerin hiçbiri gelmedi. / students-GEN none.of.them come.NEG.PST.3SG / 'None of the students came.'	[A]
21	PART	Hiçbiri gelmedi. / none.of.them come.NEG.PST.3SG / 'None came.' Context/notes: Pronounization; antecedent set salient.	[E]
22	WH	Hangi öğrenci geldi? / hangi which / öğrenci student / geldi came / 'Which student came?'	[A]
23	WH	Hangi (öğrenci) geldi? / 'Which (student) came?' Context/notes: Ellipsis candidate.	[E]
24	WH	Hangisi geldi? / hangisi which.one (pronounized) / geldi came / 'Which one came?' Context/notes: Pronounization.	[E]
25	WH	«Hangi geldi? / Intended bare selector Context/notes: Ungrammatical control.	[U]
26	WH	Kimin kitabı bu? / kimin whose / kitabı book.POSS / bu this / 'Whose book is this?'	[A]
27	WH	Kimin (kitabı) bu? / 'Whose (book) is this?' Context/notes: Ellipsis candidate.	[E]
28	WH	Kimin bu? / 'Whose is this?' Context/notes: May be acceptable as possessive pronoun question; classify support strategy.	[C]
29	STACK	Her üç öğrenci geldi. / her each / üç three / öğrenci student / geldi came / 'Every three students came.' Context/notes: Requires salient partition function π into 3-sized cells (grouping context).	[C]
30	STACK	«Üç her öğrenci geldi. / three each student came Context/notes: Ungrammatical control for Q(OP(NP)).	[U]

31	STACK	Üç öğrenci geldi. / three student came / ‘Three students came.’	[A]
32	STACK	Her üç öğrenci bir arada geldi. / each three student together came / ‘Each group of three students came together.’ Context/notes: Grouping made explicit; partition context.	[C]
33	STACK	Her üç öğrenciden biri geldi. / each three students-ABL one came / ‘One out of every three students came.’ Context/notes: Partition + selection within each cell.	[C]
34	STACK	Öğrencilerin her üçü geldi. / students-GEN each three.POSS came / Intended: ‘All three of the students came.’ Context/notes: Check whether ‘her üçü’ is lexicalized or requires discourse.	[C]
35	STACK	Her bir öğrenci geldi. / each one student came / ‘Each student came.’	[A]

10. Discussion and Conclusion

10.1 Why the split matters. The operator/quantifier split is not a terminological preference; it preserves a set of empirical generalizations that are otherwise hidden when all determiners are treated as a single class. In particular, it explains why headedness and replacement correlate with quantity meanings (quantifiers) rather than with scope meanings (operators).

10.2 Relationship to DP approaches. The manuscript’s critique of DP is best read as a critique of category uniformity rather than as a claim that determiner heads cannot exist. The architecture allows that a language may realize definiteness or deictic packaging in a high position, but it denies that all determiners, quantifiers, and operators should be conflated into a single head category.

10.3 Relationship to generalized quantifier theory. The architecture is compatible with generalized quantifier semantics for the quantifier layer (Barwise & Cooper 1981; Keenan & Stavi 1986) while proposing that some items commonly treated as generalized quantifiers are in fact operators over those quantifiers.

10.4 Summary. The three-layer architecture provides a cognitively and typologically motivated decomposition of nominal structure that separates conceptual content from quantity and from scope. It unifies headedness diagnostics, interrogative behavior, and cross-linguistic inventory facts in a compact system that is both formally typable and empirically testable.

10.5 Distinguishing predictions unique to the operator/quantifier split.

A reviewer may reasonably ask whether re-labeling every/each/no as “operators” rather than generalized quantifiers is more than terminology. The proposal is intended to be

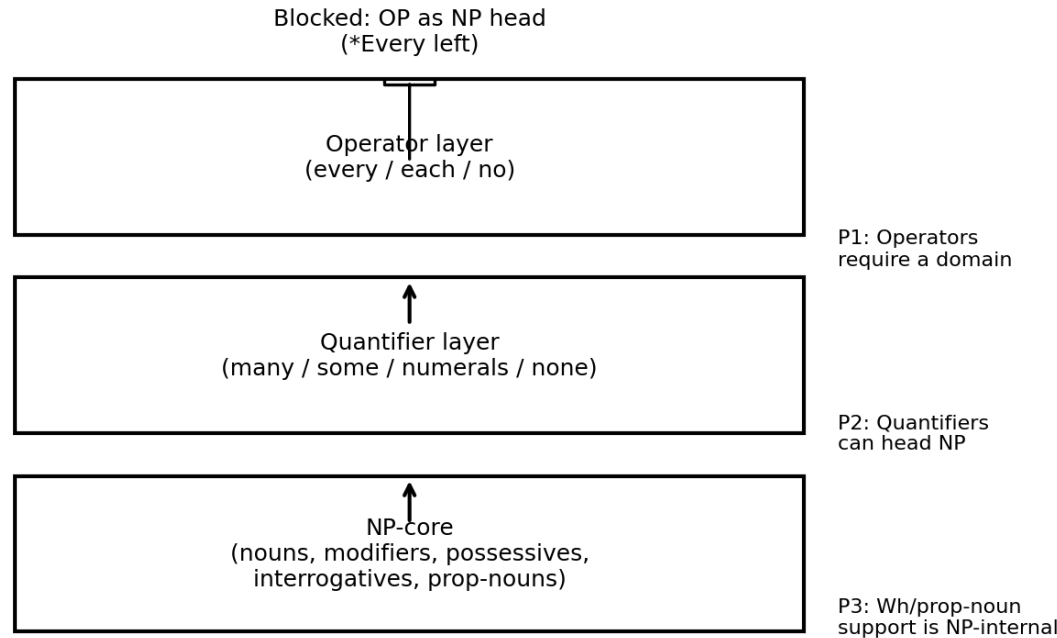
empirically committal: it predicts (i) a restricted space of grammatical stacking configurations, (ii) a systematic asymmetry in anaphora/ellipsis and partitive licensing, and (iii) cross-linguistic correlations between apparent “bare” operator uses and independently motivated NP-ellipsis/pro-drop mechanisms. These predictions follow from the typed architecture in Section 9: quantifiers deliver generalized-quantifier objects that can satisfy nominal substitution, whereas operators are scope instructions that require a domain argument (Barwise & Cooper 1981; Keenan & Stavi 1986; Xakimqulova 2024).

Prediction	Illustrative English contrast	Cross-linguistic diagnostic consequence	Why this is not a DP/GQT default
P1. Headability vs. domain-requirement split	Many left. vs. «Every left. Every student left.	Operator-alone surface forms should correlate with NP-ellipsis/pro-drop; otherwise an overt NP-core is required (Table 8).	A uniform GQ determiner class predicts similar type-availability; the asymmetry must be stipulated via extra type-shifts or silent structure.
P2. Partitive asymmetry and “prop-noun repair”	Many of the students... vs. «Every of the students... Every one of the students...	Where “of/partitive” structures exist, quantifiers should combine directly; operators require an overt nominal support strategy (prop-noun, classifier, or lexicalized ‘each-person’).	DP/GQT analyses often treat partitives as selectional quirks; here the asymmetry is derived from layer typing and headedness.
P3. Anaphora/ellipsis licensing	I read many books; you read many. «...you read every. ...you read every one.	Quantifier-headed ellipsis should be broadly available; operator-headed ellipsis is blocked unless an NP-core placeholder is present or	A GQ-only view expects both items to be of comparable semantic type; anaphora contrast then becomes construction-

		recoverable.	specific.
P4. Controlled operator–quantifier stacking	Every three students... / No many students... (scope interactions)	Only stacks that match OP(Q(NP)) (or OP(NP)) should be productive; unattested stacks indicate type mismatch rather than arbitrary word-order bans. Table 9 enumerates the predicted stacking space and status codes.	DP accounts allow multiple functional heads; without typing constraints, overgeneration is controlled by stipulations about feature compatibility.

Table 10. Distinguishing predictions that follow from treating every/each/no as operators rather than quantifier heads.

Figure 5. Layer typing and the source of the three core predictions (schematic).



10.6 What the operator category predicts beyond generalized-quantifier labeling.

Many analyses treat every/each as generalized quantifiers (GQs) because they contribute universal force. The present proposal does not dispute the truth-conditional universality;

it disputes the “layer location” and the «headedness consequences». Under the tripartite model, universality is implemented as a scope-taking operation over a domain built in NP-core (and optionally already magnitude-shaped by a quantifier). This yields three commitments that are difficult to obtain “for free” under a flat GQ determiner inventory: (i) the systematic link between universality/distributivity and non-headability, (ii) the need for explicit nominal support strategies (prop-nouns, classifier supports, lexicalized ‘each-person’ forms) when the operator is used without an overt noun, and (iii) a constrained stacking space that avoids overgeneration without proliferating DP functional heads (Barwise & Cooper 1981; Partee 1987; Szabolcsi 2010; Xakimqulova 2024).

Empirical locus	Operator-layer account (this paper)	Typical GQ-determiner account (pressure point)
Bare uses / “pronoun” behavior	Allowed for quantifiers (true headability); operator-alone requires NP-ellipsis or an overt nominal placeholder (one, classifier phrase, lexicalized ‘each-person’).	Often derives bare uses via type shifting or by treating D as independently nominal; headedness contrasts become construction-specific.
Partitives	Quantifiers select partitive complements; operators do not select ‘of’-partitives directly and instead require repair via nominal support.	Partitive selection treated as lexical idiosyncrasy; no architecture-level reason for the asymmetry.
Stacking restrictions	Only typed stacks OP(Q(NP)) and OP(NP) are predicted; failure cases indicate type mismatch.	Multiple functional heads are available; restrictions typically encoded with features or ordering constraints.
Interrogative determiners	which/whose are NP- internal domain selectors; they require NP-core («Which left) and interface with clause-level question operators.	which often treated as D/Q with alternative semantics; NP-internal dependence must be encoded separately.

Table 11. Clarifying the reviewer pressure point: what changes if every/each are operators rather than quantifier heads.

See Appendix C.1a (Table C.1a; N_items=15) for the reference-attested German pilot subset.

10.7 Summary of added commitments. The operator/quantifier split is intended to be falsifiable: if a language allows true operator headability with no independent ellipsis/pro-drop analysis, or if operator items freely select the same partitive complements as quantifiers without any nominal support strategy, the present layer typing would require revision. Conversely, the availability of repair strategies (one, classifier supports, lexicalized ‘each-person’) and the correlation between apparent operator-alone uses and ellipsis mechanisms are predicted and can be tested with the diagnostic battery in Table 8.

11. Neurolinguistic Predictions (Extension; explicitly inferred)

This section is an application-oriented extension. It takes the manuscript’s cognitive partition—content (NP-core), quantity (quantifier layer), scope (operator layer)—and proposes testable processing predictions. The aim is to provide concrete experimental hooks, not to propose a new theory of nominal structure.

11.1 Prediction: separable processing stages for domain formation, quantity, and scope. If NP-core composition corresponds to conceptual domain construction (Langacker 1987), then manipulations that increase NP-core restriction (e.g., multiple adjectives, relational modifiers) should primarily affect conceptual integration measures. By contrast, adding a quantifier should add a magnitude/proportion computation, and adding an operator should add distributive/negative scope computation. The architecture therefore predicts partial temporal and topographic separability of these effects in time-resolved measures.

11.2 ERP-oriented hypotheses. A conservative hypothesis is that NP-core conceptual integration modulates N400-like activity associated with semantic expectancy and integration (Kutas & Federmeier 2011). Operator computation, especially negation and distributive enforcement, may impose later integration/control costs that could manifest as later positivities or reanalysis-related activity, depending on task demands (Friederici 2011; Hagoort 2013). These are hypotheses rather than commitments of the theory; the key point is that the layer partition suggests where to look for dissociations.

11.3 Domain-size manipulation for interrogatives. Because which is NP-internal domain selection (Hamblin 1973; Karttunen 1977), a straightforward prediction is that increasing the size of the salient domain (e.g., visual world displays with 2 vs 20 candidate objects) increases processing cost localized to the nominal region, even when clausal structure is constant. This would be difficult to explain if which were primarily a clause-level quantifier head.

11.4 Distributivity contrasts. The operator analysis predicts that distributive enforcement should be visible when the predicate encourages collective readings but the operator forces distributive readings. Minimal pairs can be built where only operator

status changes (e.g., “Many students lifted the piano” vs “Each student lifted the piano”), predicting different processing patterns attributable to distributive scope rather than to magnitude.

11.5 Suggested experimental designs. (i) Eye-tracking/self-paced reading: cross NP-core complexity (simple noun vs noun with multiple restrictors) with quantifier presence (many/three/none) and operator presence (each/every/no). (ii) EEG/MEG: factorial designs isolating domain restriction (which) from quantifier magnitude and operator scope. (iii) Behavioral verification: picture verification tasks comparing operator-driven distributivity vs quantifier-driven magnitude.

11.6 Neural localization (cautious). The architecture does not require a one-to-one mapping from layer to brain region. Nevertheless, unification-oriented views of language suggest that different computations (lexico-conceptual integration vs scope/control) may engage partially different networks (Hagoort 2013). The layer partition provides a principled way to define computational targets for such studies.

Figure 4. Hypothesized processing sequence for nominal layers (prediction/inference section)

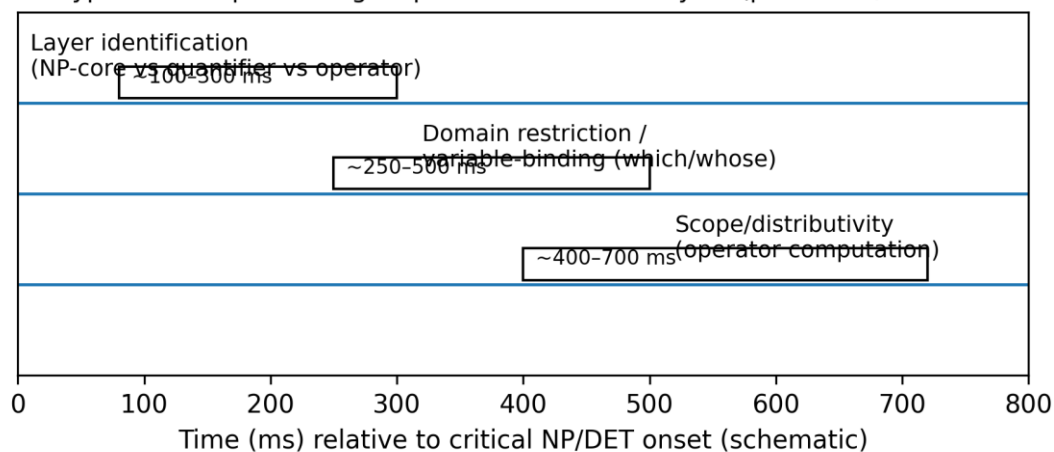


Figure 6. Hypothesized processing sequence for nominal layers (prediction/inference section).

12. Computational Applications (Extension; explicitly inferred)

This section is a second application-oriented extension. It treats the manuscript’s layer partition as a representational choice that can reduce ambiguity in NLP systems: specifically, it helps prevent conflating scope operators (every/each/no) with magnitude quantifiers (many/some/few) during semantic assembly.

12.1 Why current NLP systems confuse operators and quantifiers. In many pipelines, determiners are treated as function words with limited semantic contribution, while quantificational meaning is approximated through heuristics or learned implicitly. This makes it difficult to diagnose errors where a model mishandles distributivity (each/every)

or negation (no). The manuscript’s architecture recommends explicit layer labeling as a lightweight interpretability scaffold.

12.2 Annotation schema derived from the manuscript. A minimal annotation scheme assigns the following labels: NP-CORE span, QHEAD (quantifier head) with QCOMP (its NP-core complement), OP (operator) with OP-SCOPE (the span it scopes over), and WH (interrogative operator) with WH-DOMAIN (NP-core restriction). The scheme is intentionally small and corresponds directly to the diagnostics in Table 2.

12.3 Rule-based baseline. As a baseline, a deterministic classifier can implement the headedness and replacement diagnostics on shallow parses: closed-class operator inventory (each/every/no), closed-class wh inventory (which/whose), and a quantifier lexicon including numerals and proportional items. Ambiguous items (e.g., one) are resolved by local syntactic context: prop-noun one occurs with determiners/adjectives as a nominal head; numeral one occurs as a quantifier selecting an NP complement.

12.4 Typed semantic assembly. The formal appendix already supplies a type ladder. In a computational implementation, NP-core yields a set-denoting predicate $A(x)$. Quantifiers yield generalized quantifier objects $Q(A)$. Operators yield scope-taking constraints over these objects. Interrogatives yield domain restrictions that can be passed to a question operator at the clause level. The benefit is modularity: errors can be located to the layer where they occur.

12.5 Downstream tasks. The layer annotations can be used to improve question answering (distinguishing “Which book...” domain restriction from “How many books...” magnitude), information extraction (negation scope under no), and natural language inference (distinguishing distributive entailments under each/every from proportional entailments under many/most).

12.6 Evaluation: minimal proof of usefulness (toy plan). To make the computational section publishable, the proposal can be evaluated with a small, diagnostic-aligned benchmark: (i) annotate a pilot set of ~300 English sentences plus ~100 per language (German/Mandarin/Turkish) using the layer tags (NP-CORE, Q, OP, WH, ONE-support) and the judgment labels [A/M/C/E]; (ii) report inter-annotator agreement targets (e.g., Cohen’s $\kappa \geq 0.75$ for layer tags; $\kappa \geq 0.80$ for [A/M/C/E] on the diagnostic suite); (iii) construct a synthetic/controlled test suite by crossing five diagnostics (headedness, replacement, partitives, WH-support, stacking) with 10–20 lexical instantiations per language; (iv) compare a baseline model that uses standard POS/UD features (DET/PRON/NUM) to an ablation model that additionally uses the layer tags. The core claim is not that the layer tags solve general parsing, but that they measurably improve prediction of the diagnostic outcomes (especially operator-alone acceptability and repair strategies) relative to POS-only baselines.

Concrete measurable endpoints (recommended). Primary metric: accuracy/F1 on predicting diagnostic labels for the targeted suite (Table 8 mini-sets plus additional items). Secondary metrics: (a) confusion reduction between OP vs Q in bare uses, (b) correct prediction of required repairs (one/classifier/her bir), and (c) agreement between predicted and attested stacking statuses (Table 9).

12.7 Practical integration. The layer tags can be integrated as features in structured prediction models, or used to supervise a neural encoder. Even when end-to-end models are used, explicit supervision on layer distinctions supports interpretability and more reliable generalization.

Figure 5. Implementation-oriented pipeline derived from the paper’s layer distinctions

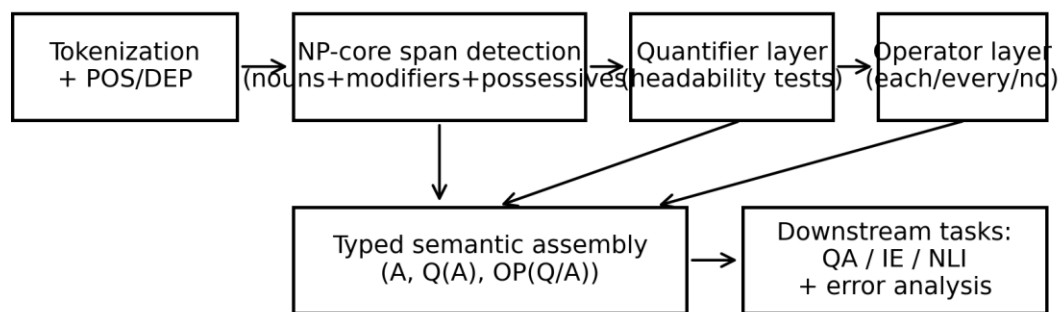


Figure 7. Implementation-oriented pipeline derived from the paper’s layer distinctions.

13. Limitations and Future Work

First, the cross-linguistic comparison covers four languages chosen for typological diversity, but broader sampling is needed to refine how the layers are realized in languages with richer classifier systems, with determiner doubling, or with freer word order (Haspelmath 1997; Dryer 2013).

Second, the operator inventory is intentionally narrow (every/each/no). Future work should test additional candidates (e.g., distributive markers, negative polarity determiners) and determine whether they belong to the operator layer or require a finer sub-typing of operator meanings.

Third, the manuscript treats articles/demonstratives as NP-internal packaging elements *distinct* from quantifiers and operators. A future expansion could formalize

definiteness/deixis more directly and test how these items interact with scope in languages with determiner spreading.

Fourth, the extension sections propose neurolinguistic and computational applications as inferences. These are offered as concrete next steps; they require empirical validation and careful experimental design.

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